

DNA Transcription & Translation Practice Test

1. For the given phrase, choose the type of molecule that is most closely associated with the phrase.

contains the hereditary information passed on from generation to generation in humans

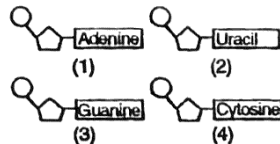
- A) DNA, only
B) messenger RNA, only
C) transfer RNA, only
D) messenger and transfer RNA, only

2. Examples of polymers which contain repeating units known as nucleotides are

- A) DNA and RNA
B) hemoglobin and maltase
C) fats and oils
D) starch and glycogen

Questions 3 and 4 refer to the following:

The diagrams below represent nucleotides.



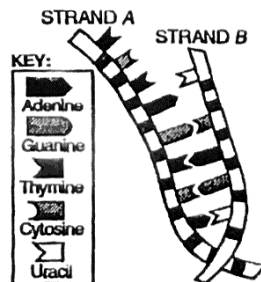
3. Which pair of nucleotides can be held together by weak hydrogen bonds?

- A) 2 and 3
B) 1 and 3
C) 4 and 2
D) 3 and 4

4. Which structure pairs with the adenine nucleotide during the formation of messenger RNA?

- A) 1
B) 2
C) 3
D) 4

Questions 5 through 7 refer to the following:



5. If the diagram represents a part of the process of protein synthesis, strand A would

- A) pick up and transfer nucleic acids to the nucleus
B) pick up and transfer specific amino acids to the cytoplasm
C) serve as a template for the synthesis of messenger RNA
D) carry a code determined by the original DNA molecule from the nucleus to the cytoplasm

6. If strand A represents a portion of a DNA molecule, its complementary sequence of nitrogenous bases on messenger RNA would normally be

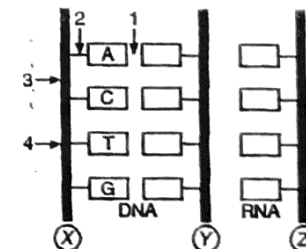
- A) T-C-T-A-G-T-C-T
B) U-G-U-A-G-U-C-U
C) A-G-A-T-C-A-G-T
D) A-G-A-U-C-A-G-U

7. Strand A would normally be found in the

- A) ribosome
B) nucleus
C) plasma membrane
D) vacuole

Questions 8 through 10 refer to the following:

The diagram below represents parts of two nucleic acid molecules.



8. What is the normal nitrogenous base sequence in the segment of strand Y shown in the diagram?

- A) T-G-U-C
B) U-G-A-C
C) T-G-A-C
D) T-A-A-C

9. Molecules represented by strand Z move from the nucleus of a cell to the cytoplasmic organelles known as

- A) centrioles
B) vacuoles
C) ribosomes
D) mitochondria

10. If strand X serves as a template for the synthesis of strand Z, the base sequence of that fragment of strand Z shown is

- A) A-C-U-G
B) U-G-A-C
C) A-C-T-G
D) T-G-A-C

11. Which nitrogenous bases tend to pair with each other in a double-stranded molecule of DNA?

- A) thymine-adenine
B) guanine-adenine
C) cytosine-thymine
D) adenine-uracil

12. Two types of RNA molecules are

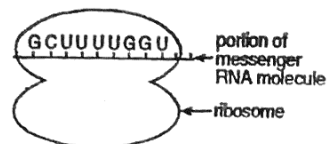
- A) messenger RNA and transfer RNA
B) uracil and adenine
C) cytosine and thymine
D) transfer RNA and translocation RNA

13. The function of transfer RNA molecules is to

- A) transport amino acids to messenger RNA
B) provide a template for the synthesis of messenger RNA
C) synthesize more transfer RNA molecules
D) transport amino acids to DNA in the nucleus

Questions 14 through 18 refer to the following:

The diagram below represents a portion of a messenger RNA molecule associated with a ribosome.



14. The presence of which nitrogen base indicates that the molecule associated with the ribosome is RNA?
- A) adenine B) guanine C) cytosine D) uracil

15. The sequence of nucleotides on the RNA molecule was determined by the
- A) sequence of nucleotides on transfer RNA molecules
B) base sequence of the original messenger RNA molecule that served as a template
C) base sequence of the original DNA molecule that served as the template
D) sequence of amino acids that will be linked together to form a polypeptide chain

16. The messenger RNA genetic codes for 3 different amino acids are:

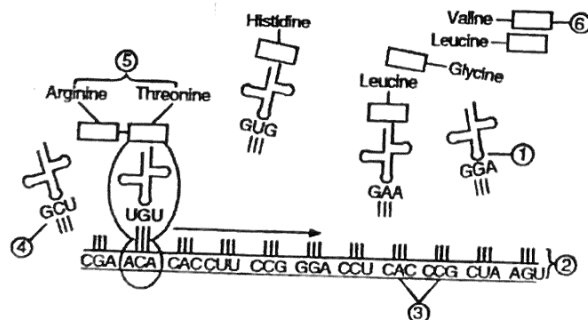
U-U-U = phenylalanine
G-C-U = alanine
G-G-U = glycine

Using this information, the strip of messenger RNA shown in the illustration would result in an amino acid sequence consisting of

- A) alanine — phenylalanine — glycine C) phenylalanine — alanine — glycine
B) alanine — glycine — glycine D) alanine — glycine — phenylalanine
17. The association between the ribosome and the messenger RNA molecule occurs in the
- A) cytoplasm B) nucleolus C) nucleus D) centrosome
18. The substance being synthesized in the cell is most likely a
- A) fat B) polypeptide C) vitamin D) carbohydrate

Questions 19 through 22 refer to the following:

The diagram below illustrates protein synthesis.



- 19) In the diagram, A, C, G, and U represent
- A) amino acids B) ribosomes C) fatty acids D) nitrogenous bases

- 20) Which structures are composed of RNA?
- A) 5 and 6 B) 4 and 5 C) 1 and 2 D) 3 and 4

- 21) The portion of DNA that carries the code for leucine is
- A) C-T-T B) G-T-T C) G-A-A D) C-U-U
- 22) The process that results in the formation of structure 2 is
- A) RNA synthesis C) crossing-over
B) nondisjunction D) hydrolysis

- 23) For the given phrase, choose the type of molecule that is most closely associated with the phrase

may contain the nitrogenous base uracil

- A) transfer RNA, only C) messenger and transfer RNA, only
B) messenger RNA, only D) DNA, only

- 24) For the given phrase, choose the cell compound which is best described by the phrase.

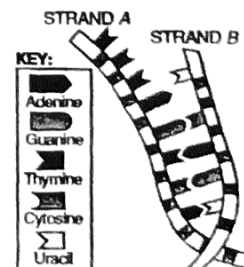
Carries protein building blocks to ribosomes.

- A) transfer RNA C) ATP
B) messenger RNA D) DNA

- 25) The genetic code for one amino acid molecule consists of

- A) three nucleotides C) four hydrogen bonds
B) two phosphates D) five sugar molecules

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26)

If strand B represents messenger RNA, it would transport the genetic code from the

- A) nucleus to the ribosome C) ribosome to the nucleus
B) mitochondria to the nucleus D) nucleus to the mitochondria

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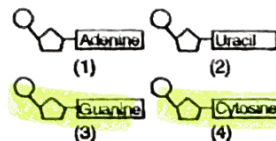
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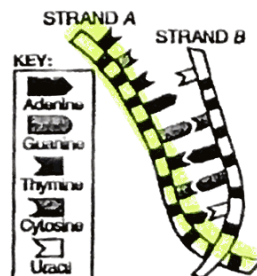
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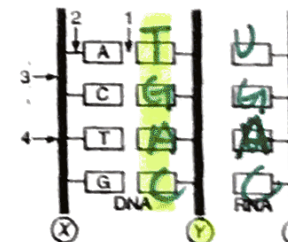


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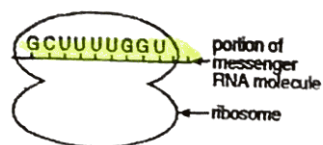
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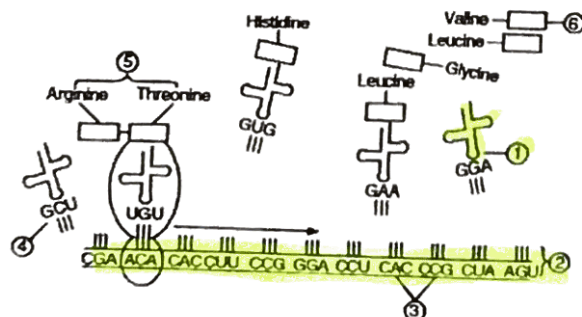
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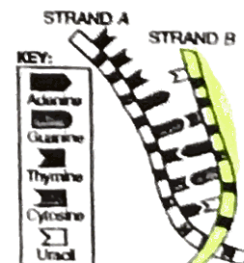
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